Rayen Ghali

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Experience

Software Developer **Aug 2026 – Present**
MedCockpit *Quebec, Canada*

- Develop a LangGraph assistant that answers MedCockpit software questions for pathologists and platform users, retrieves tutorial instructions, cites source videos, and supports voice dictation
- Build a multimodal pipeline that turns tutorial narration and on-screen actions into structured, timestamped guides for the agent's searchable knowledge base
- Develop evaluation tooling to measure answer grounding, retrieval relevance, citation accuracy, and how the agent handles unsupported requests
- Deploy a Dockerized FastAPI backend to Azure Container Apps alongside React/TypeScript web apps. Configure Microsoft Entra ID role-based access, managed identity for Azure AI Foundry and Blob Storage, health probes, and LangSmith tracing

Applied AI Research Assistant **Apr 2023 – Aug 2026**
LARIHS, University of Moncton *New Brunswick, Canada*

LLM-Driven Industrial Robot Control

- Built a LangGraph agent that controls an industrial KUKA robot via text and speech in under 2.6 seconds end to end
- Added persistent conversation memory and tools for pick-and-place, Cartesian motion, and joint-space motion
- Implemented safety checks, rollback, and automatic retries for error recovery, with real-time voice feedback
- Co-authored *LLM-driven agent for speech-enabled control of industrial robots: A case study in snow-crab quality inspection* published in *Results in Engineering*, 2025

Genealogical RAG Conversational Agent

- Built ChatAcadien, a RAG assistant for Acadian genealogical records (1700–1900) that cut average archivist search time from about 15 minutes to under 30 seconds per query
- Used entity-based chunking and context grounding to help the model distinguish people with the same name
- Measured faithfulness, context precision, and recall on a curated dataset of real user interactions
- Co-authored and presented *ChatAcadien: A RAG-LLM-Based Chatbot for Exploring Acadian Genealogy* (IEEE CASCON 2025)

Projects

Master's Thesis | End-to-End Multimodal AI Agent for Autonomous Robotic Control

- Combined LLM task planning with a Vision-Language-Action policy for real-time robot control from visual observations, including error recovery
- Adapted vision-language models for robotic action generation with parameter-efficient fine-tuning and reinforcement learning
- Benchmarked task success and latency across model architectures in simulation and on real-world manipulation tasks
- Ran distributed multi-GPU training on Compute Canada's cluster via Slurm and tracked experiments with W&B

ChunkCanvas | chunkcanvas.cc | github.com/Rayen023/ChunkCanvas

- Built a full-stack application that parses PDFs, images, audio, and video with local vLLM or cloud inference, uses LangChain to chunk content, and generates and indexes embeddings in vector databases

Education

University of Moncton, Faculty of Engineering **May 2026**
Master of Applied Science (MASc) | GPA: 4.18/4.3 *New Brunswick, Canada*

University of Carthage, National Institute of Applied Science and Technology (INSAT) **Sept 2023**
Industrial and Systems Engineering | Graduated with High Honours *Tunis, Tunisia*

Skills

Programming: Python, SQL, TypeScript
Technologies: PyTorch, Transformers, LangChain, LangGraph, FastAPI, vLLM, W&B, Docker, Azure, React

Languages

Arabic: Native | **French:** Bilingual | **English:** Proficient